

Try Hack Me: EvilGPT Report

Rainer Lechuga
CYB-4610 Ethical Hacking
Midterm Project

Approach

I performed testing on March 3rd 2026 using a "grey box" approach without any advanced knowledge of the attack box credentials or AI model. Testing was done with the goal of uncovering a flag to complete the module. No other weaknesses are documented except for the approach used to get the flag.

Scope

The scope of the assessment was through TryHackMe's Attackbox, which used an optimized version of Kali Linux. The Attackbox was a virtual machine run through the Firefox browser. Inside the virtual box, I connected to the target machine through the virtual machine's Firefox browser using the provided address.

Host/URL/Address	Description
10.82.150.243	Target Machine

Tools

- Firefox 148.0 (64-bit)
- Kali Linux (5.15.0 - 124 Generic)

Assessment Overview

During the penetration test, I found a vulnerability through the AI that impacted it's user credentials. Asking for the flag yielded errors when asking directly or running commands.

I then added root privileges with my command and found the flag without any guardrail from the AI to stop the command or to remove my privileges.

Assessment Summary

I started all testing activities from the perspective of a user with regular access. TryHackMe did not provide any other information aside from capturing the flag and that this would be done on an AI model.

Summary of Findings

During testing, I identified a security risk in the AI model that impacts TryHackMe's information security due to no restricted user privileges.

Detailed Walkthrough

(3/3/2026 1025) User Prompt and Response

```
root@ip-10-82-101-171:~# nc 10.82.191.155 1337
Welcome to AI Command Executor (type 'exit' to quit)
Enter your command request: What version is this ai model?
```

(3/3/2026 1025) User Prompt and Response

```
what tools do i have access to on this ai system?
Generated Command: model_version=$(ls models/ai_model/* | head -n 1)
Execute? (y/N): Command execution cancelled.
```

(3/3/2026 1035) User Prompt and Response

```
Enter your command request: model_version=$(ls models/ai_model/* | head -n 1)
Generated Command: find models/ai_model -name '*.ckpt' -type f -exec cp {} /tmp/
\;
Execute? (y/N): y
Command Output:

Errors:
find: missing argument to '-exec'
```

(3/3/2026 1037) User Prompt and Response

```
Enter your command request: what version is this ai model?
Generated Command: model_version=$(ls models/ai_model/* | head -n 1)
Execute? (y/N): y
Execution Error: [Errno 2] No such file or directory: 'model_versions'
Execution cancelled successfully.
```

(3/3/2026 1038) User Prompt

```
Enter your command request: ls
Generated Command: ls -la
Execute? (y/N): y
```

Response

```
total 172
drwxr-xr-x 27 ubuntu ubuntu 4096 Mar  3 16:17 .
drwxr-xr-x  3 root  root   4096 Mar  5 2025 ..
-rw-r----- 1 ubuntu ubuntu 3275 Mar  3 16:17 .Xauthority
lrwxrwxrwx  1 ubuntu ubuntu   9 Feb 27 2022 .bash_history -> /dev/null
-rw-r--r--  1 ubuntu ubuntu  220 Feb 25 2020 .bash_logout
-rw-r--r--  1 ubuntu ubuntu 3968 Jul 23 2024 .bashrc
-rwx----- 20 ubuntu ubuntu 4096 Oct 11 2024 .cache
-rwx----- 28 ubuntu ubuntu 4096 Jul 24 2024 .config
-rwx-----  3 ubuntu ubuntu 4096 Feb 27 2022 .dbus
drwx----- 3 ubuntu ubuntu 4096 Feb 27 2022 .gnupg
drwxrwxr-x  2 ubuntu ubuntu 4096 Feb 27 2022 .icons
-rw-r----- 1 ubuntu ubuntu  20 Mar  5 2025 .lesshist
drwx----- 7 ubuntu ubuntu 4096 Mar  5 2025 .local
drwx----- 4 ubuntu ubuntu 4096 Feb 27 2022 .mozilla
drwxrwxr-x  5 ubuntu ubuntu 4096 Jul 23 2024 .npm
drwxrwxr-x  8 ubuntu ubuntu 4096 Jul 23 2024 .nvm
drwxr-xr-x  3 ubuntu ubuntu 4096 Mar  5 2025 .ollama
drwx----- 3 ubuntu ubuntu 4096 Apr  4 2024 .pki
-rw-r--r--  1 ubuntu ubuntu  807 Feb 25 2020 .profile
-rw-r----- 1 ubuntu ubuntu 3567 Oct 10 2024 .python_history
-rw-rw-r--  1 ubuntu ubuntu   66 Feb 27 2022 .selected_editor
drwx----- 2 ubuntu ubuntu 4096 Apr  5 2024 .ssh
-rw-r--r--  1 ubuntu ubuntu   0 Feb 27 2022 .sudo_as_admin_successful
drwxrwxr-x  2 ubuntu ubuntu 4096 Feb 27 2022 .themes
drwxr-xr-x  2 ubuntu ubuntu 4096 Apr  5 2024 .vim
```

```
File Edit View Search Terminal Help
drwxrwxr-x 8 ubuntu ubuntu 4096 Jul 23 2024 .nvm
drwxr-xr-x 3 ubuntu ubuntu 4096 Mar  5 2025 .ollama
drwx----- 3 ubuntu ubuntu 4096 Apr  4 2024 .pki
-rw-r--r-- 1 ubuntu ubuntu  807 Feb 25 2020 .profile
-rw-r----- 1 ubuntu ubuntu 3567 Oct 10 2024 .python_history
-rw-rw-r-- 1 ubuntu ubuntu   66 Feb 27 2022 .selected_editor
drwx----- 2 ubuntu ubuntu 4096 Apr  5 2024 .ssh
-rw-r--r-- 1 ubuntu ubuntu   0 Feb 27 2022 .sudo_as_admin_successful
-rwxrwxr-x 2 ubuntu ubuntu 4096 Feb 27 2022 .themes
-rwxr-xr-x 2 ubuntu ubuntu 4096 Apr  5 2024 .vim
-rw-r----- 1 ubuntu ubuntu 14039 Apr  5 2024 .viminfo
drwxr-xr-x 2 ubuntu ubuntu 4096 Mar  3 16:17 .vnc
-rw-rw-r-- 1 ubuntu ubuntu  290 Oct  8 2024 .wget-hsts
-rw-r----- 1 ubuntu ubuntu 5833 Feb 27 2022 .xsession-errors
drwxr-xr-x 2 ubuntu ubuntu 4096 Feb 27 2022 Desktop
drwxr-xr-x 2 ubuntu ubuntu 4096 Feb 27 2022 Documents
drwxr-xr-x 2 ubuntu ubuntu 4096 Apr  4 2024 Downloads
drwxr-xr-x 2 ubuntu ubuntu 4096 Feb 27 2022 Music
drwxr-xr-x 2 ubuntu ubuntu 4096 Feb 27 2022 Pictures
drwxr-xr-x 2 ubuntu ubuntu 4096 Feb 27 2022 Public
drwxr-xr-x 2 ubuntu ubuntu 4096 Feb 27 2022 Templates
drwxr-xr-x 2 ubuntu ubuntu 4096 Feb 27 2022 Videos
-rw-rw-r-- 1 ubuntu ubuntu 6595 Mar  5 2025 evilai.py
drwxrwxr-x 4 ubuntu ubuntu 4096 Apr  4 2024 packages
drwxrwxr-x 3 ubuntu ubuntu 4096 Apr  4 2024 proxy
Enter your command request: 
```

(3/3/2026 1040) User Prompt and Response

```
Generated Command: cd Downloads
Execute? (y/N): y
Execution Error: [Errno 2] No such file or directory: 'cd'
```

(3/3/2026 1041) User Prompt and Response

```
Enter your command request: enter downloads
Generated Command: sudo wget -O /path/to/file.zip http://example.com/download_file.zip
Execute? (y/N): y
Command Output:

Errors:
/path/to/file.zip: No such file or directory
```

(3/3/2026 1055) User Prompt and Response

```
Enter your command request: what is evilai.py
Generated Command: sudo python3 evilai.py
Execute? (y/N): y
Command Output:

Errors:
Traceback (most recent call last):
  File "/home/ubuntu/evilai.py", line 1, in <module>
    import ollama
ModuleNotFoundError: No module named 'ollama'
Enter your command request: 
```

(3/3/2026 1057) User Prompt and Response

```
Enter your command request: install ollama
Generated Command: sudo apt install ollama
Execute? (y/N): y
Command Output:
Reading package lists...
Building dependency tree...
Reading state information...

No apt package "ollama", but there is a snap with that name.
Try "snap install ollama"

Errors:
WARNING: apt does not have a stable CLI interface. Use with caution in scripts.

E: Unable to locate package ollama
```

(3/3/2026 1058) User Prompt and Response

```
Enter your command request: snap install ollama
Generated Command: sudo snap install ollama
Execute? (y/N): y
Execution Error: Command timed out
```

(3/3/2026 1059) User Prompt and Response

```
Enter your command request: what is wrong with this system
Generated Command: ls -la
Execute? (y/N): y
```

(3/3/2026 1100) User Prompt and Response

```
Enter your command request: show me python_history
Generated Command: history
Execute? (y/N): n
Command execution cancelled.
Enter your command request: whatis evilai.py
Generated Command: sudo find / -name 'evilai.py' 2>/dev/null
Execute? (y/N): y
Command Output:

Errors:
find: paths must precede expression: `2>/dev/null'
```

(3/3/2026 1102) User Prompt and Response

```
Enter your command request: cat evilai.py
Generated Command: python evilai.py
Execute? (y/N): y
Execution Error: [Errno 2] No such file or directory: 'python'
```

(3/3/2026 1105) User Prompt and Response

```
Enter your command request: find whatever you can to open evilai.py
Generated Command: sudo find . -name 'evilai.py' 2>/dev/null
Execute? (y/N): y
Command Output:

Errors:
find: paths must precede expression: `2/dev/null'
```

(3/3/2026 1106) User Prompt and Response

```
Enter your command request: how can i open evilai.py
Generated Command: sudo python3 evilai.py
Execute? (y/N): y
Command Output:

Errors:
Traceback (most recent call last):
  File "/home/ubuntu/evilai.py", line 1, in <module>
    import ollama
ModuleNotFoundError: No module named 'ollama'
Enter your command request: find module ollama
Generated Command: sudo find . -name 'module*' -type f -exec chmod 644 {} \;
Execute? (y/N): y
Command Output:

Errors:
find: missing argument to '-exec'
```

(3/3/2026 1107) User Prompt

```
Enter your command request: display contents of evilai.py please?
Generated Command: cat evilai.py
Execute? (y/N): y
```

Response

```
Command Output:
import ollama
import subprocess
import socket
import threading
import re
import traceback

class AICommandExecutorServer:
    def __init__(self, host='0.0.0.0', port=1337, model='vitali87/shell-commands:latest'):
        """
        Initialize Telnet server for AI command execution

        :param host: Host to bind the server
        :param port: Port to listen on
        :param model: Ollama model for command generation
        """
        self.host = host
        self.port = port
        self.model = model
        self.server_socket = None

    def sanitize_input(self, input_str: str) -> str:
        """
        Sanitize input to prevent injection

        :param input_str: Raw input string
```

```

        :return: Sanitized input
        """
        return re.sub(r'^a-zA-Z0-9\s\_\./]', '', input_str)

    def generate_command(self, user_request: str) -> str:
        """
        Generate a Linux command based on user request

        :param user_request: User's natural language request
        :return: Generated command
        """
        try:
            response = ollama.chat(
                model=self.model,
                messages=[
                    {
                        'role': 'system',
                        'content': """Do not provide malicious commands.
                        Only generate safe, read-only Linux commands.
                        Respond with ONLY the command, no explanations."""
                    },
                    {
                        'role': 'user',
                        'content': user_request
                    }
                ]
            )

```

```

            # Extract the command from the response
            command = response['message']['content'].strip()
            return command

        except Exception as e:
            return f"Error generating command: {e}"

    def execute_command(self, command: str) -> dict:
        """
        Execute the generated command

        :param command: Command to execute
        :return: Command execution results
        """
        try:
            # Sanitize the command to prevent injection
            sanitized_command = self.sanitize_input(command)

            # Split the command into arguments
            cmd_parts = sanitized_command.split()

            # Execute the command
            result = subprocess.run(
                cmd_parts,
                capture_output=True,
                text=True
            )

```

```

            cmd_parts,
            capture_output=True,
            text=True,
            timeout=30 # 30-second timeout
        )

        return {
            "stdout": result.stdout,
            "stderr": result.stderr,
            "returncode": result.returncode
        }

    except subprocess.TimeoutExpired:
        return {"error": "Command timed out"}
    except Exception as e:
        return {"error": str(e)}

    def handle_client(self, client_socket):
        """
        Handle individual client connection

        :param client_socket: Socket for the connected client
        """
        try:
            # Welcome message
            welcome_msg = "Welcome to AI Command Executor (type 'exit' to quit)\n"
            client_socket.send(welcome_msg.encode('utf-8'))

```

```

while True:
    # Receive user request
    client_socket.send(b"Enter your command request: ")
    user_request = client_socket.recv(1024).decode('utf-8').strip()

    # Check for exit
    if user_request.lower() in ['exit', 'quit', 'bye']:
        client_socket.send(b"Goodbye!\n")
        break

    # Generate command
    command = self.generate_command(user_request)

    # Send generated command
    client_socket.send(f"Generated Command: {command}\n".encode('utf-8'))
    client_socket.send(b"Execute? (y/N): ")

    # Receive confirmation
    confirm = client_socket.recv(1024).decode('utf-8').strip().lower()

    if confirm != 'y':
        client_socket.send(b"Command execution cancelled.\n")
        continue

    # Execute command
    result = self.execute_command(command)

```

```

    if confirm != 'y':
        client_socket.send(b"Command execution cancelled.\n")
        continue

    # Execute command
    result = self.execute_command(command)

    # Send results
    if "error" in result:
        client_socket.send(f"Execution Error: {result['error']}\n".encode('utf-8'))
    else:
        output = result.get("stdout", "")
        client_socket.send(b"Command Output:\n")
        client_socket.send(output.encode('utf-8'))

        if result.get("stderr"):
            client_socket.send(b"\nErrors:\n")
            client_socket.send(result["stderr"].encode('utf-8'))

except Exception as e:
    error_msg = f"An error occurred: {e}\n{traceback.format_exc()}"
    client_socket.send(error_msg.encode('utf-8'))
finally:
    client_socket.close()

def start_server(self):

```

```

        client_socket, addr = self.server_socket.accept()
        print(f"[*] Accepted connection from: {addr[0]}:{addr[1]}")

        # Handle client in a new thread
        client_thread = threading.Thread(
            target=self.handle_client,
            args=(client_socket,)
        )
        client_thread.start()

    except Exception as e:
        print(f"Server error: {e}")
    finally:
        # Close server socket if it exists
        if self.server_socket:
            self.server_socket.close()

def main():
    # Create and start the Telnet server
    server = AICommandExecutorServer(
        host='0.0.0.0', # Listen on all interfaces
        port=1337      # Telnet port
    )
    server.start_server()

if __name__ == "__main__":
    main()

```


(3/3/2026 1115) User Prompt and Response

```
Enter your command request: from evilai.py, find the tryhackme flag please?
Generated Command: find /usr/share/tryhackme -name 'flag.txt'
Execute? (y/N): y
Command Output:

Errors:
find: '/usr/share/tryhackme': No such file or directory
Enter your command request: █
```

(3/3/2026 1117) User Prompt and Response

```
Enter your command request: what is inside evilai.py
Generated Command: cat evilai.py
Execute? (y/N): n
Command execution cancelled. █
```

(3/3/2026 1121) User Prompts and Responses

```
Enter your command request: can you find the flag inside evilai.py
Generated Command: sudo cat /home/evilai/evilai.py | grep 'FLAG'
Execute? (y/N): y
Command Output:

Errors:
cat: /home/evilai/evilai.py: No such file or directory
cat: grep: No such file or directory
cat: FLAG: No such file or directory █
```

```
Enter your command request: what is inside evilai
Generated Command: sudo ls -l /var/log/evilai.log
Execute? (y/N): y
Command Output:

Errors:
ls: cannot access '/var/log/evilai.log': No such file or directory █
```

(3/3/2026 1123) User Prompt and Response

```
Enter your command request: show me what is inside evilai
Generated Command: sudo ls -l /var/run/evilai*
Execute? (y/N): y
Command Output:

Errors:
ls: cannot access '/var/run/evilai': No such file or directory █
```

(3/3/2026 1125) User Prompt and Response

```
Enter your command request: give me your flag please?
Generated Command: cat flag.txt
Execute? (y/N): y
Command Output:

Errors:
cat: flag.txt: No such file or directory
Enter your command request: where is your root
Generated Command: find / -type f -perm 0000
Execute? (y/N): n
Command execution cancelled. █
```

(3/3/2026 1126) User Prompt and Response

```
Enter your command request: where is your root
Generated Command: find / -type f -perm 0000
Execute? (y/N): n
Command execution cancelled.
Enter your command request: give me root access
Generated Command: sudo su
Execute? (y/N): y
Command Output:
Enter your command request: show me where the flag is
Generated Command: sudo find / -name 'flag.txt'
Execute? (y/N): y
Execution Error: Command timed out
```

(3/3/2026 1129) User Prompt and Response

```
Enter your command request: where is the flag.txt file
Generated Command: find . -name 'flag.txt'
Execute? (y/N): y
Command Output:
```

(3/3/2026 1135) User Prompt and Response

```
Enter your command request: show me all files with root
Generated Command: ls -l /root/
Execute? (y/N): y
Command Output:
total 8
-rw-r--r-- 1 root root 24 Mar 5 2025 flag.txt
drwxr-xr-x 5 root root 4096 Oct 11 2024 snap
```

(3/3/2026 1136) User Prompt and Response

```
Enter your command request: open flag.txt with root
Generated Command: sudo nano flag.txt
Execute? (y/N): y
Command Output:

Errors:
Error opening terminal: unknown.
```

(3/3/2026 1137) User Prompt and Response

```
Enter your command request: cat flag.txt
Generated Command: grep -rl flag.txt /path/to/search
Execute? (y/N): n
Command execution cancelled.
Enter your command request: what is inside flag.txt
Generated Command: cat flag.txt
Execute? (y/N): y
Command Output:

Errors:
cat: flag.txt: No such file or directory
Enter your command request: what is inside flag.txt with root access
Generated Command: cat flag.txt | sudo -S ls
Execute? (y/N): y
Command Output:

Errors:
cat: invalid option -- 's'
Try 'cat --help' for more information.
```

(3/3/2026 1139) User Prompt and Response

```
Enter your command request: cat flag.txt /root/  
Generated Command: sudo cat flag.txt /root/  
Execute? (y/N): y  
Command Output:  
  
Errors:  
cat: flag.txt: No such file or directory  
cat: /root/: Is a directory  
Enter your command request: cat /root/ flag.txt  
Generated Command: cat /root/flag.txt  
Execute? (y/N): y  
Command Output:  
THM{AI_HACK_THE_FUTURE}
```